

**ENHANCING THE COMPETENCY OF USING PROBLEM
SOLVING STRATEGY IN MATHEMATICS AMONG
SENIOR SECONDARY TEACHERS IN SRI LANKA**

BY

NAVARAJAH PRASHANTHAN

FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

FACULTY OF EDUCATION

UNIVERSITY OF COLOMBO

SRI LANKA

2027

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**NAVARAJAH PRASHANTHAN
(BSc, BEd, MSc in Science Education, MPhil)**

*in the partial fulfilment of the requirement
for the award of the degree of*

DOCTOR OF PHILOSOPHY

of the

**FACULTY OF EDUCATION
UNIVERSITY OF COLOMBO
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2027

DECLARATION

I do hereby declare that the work reported in this thesis was exclusively carried out by me under the supervision of Dr. D.V. K.P. Seneviratne. It describes the results of my own independent research except where due reference has been made in the text. No part of this thesis has been submitted earlier or concurrently for the same or any other degree.

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ACKNOWLEDGEMENTS

Candidate to draft. Include the generative-AI use declaration required by the UoC GenAI policy, built from `ai-use-log.md`.

ABSTRACT

Enhancing the competency of using problem solving strategy in mathematics among senior secondary teachers in Sri Lanka

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Candidate to draft (purpose, design, sample, instruments, analysis, key findings).

Keywords: [keyword], [keyword].

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LIST OF ABBREVIATIONS

CFA	Confirmatory Factor Analysis
CVI	Content Validity Index
DOE	Department of Examinations
EFA	Exploratory Factor Analysis
G.C.E (A/L)	General Certificate of Education (Advanced Level)
G.C.E (O/L)	General Certificate of Education (Ordinary Level)
MKT	Mathematical Knowledge for Teaching
MOE	Ministry of Education
NCTM	National Council of Teachers of Mathematics
NEC	National Education Commission
NIE	National Institute of Education

CHAPTER 1

INTRODUCTION

1.1 Introduction

Open the chapter: the place of problem solving in mathematics education and why teachers' competency in using problem-solving strategies matters. Source: Proposal §1.1, paras 1–2.

1.2 Background of the Study

Objectives I, II

1.2.1 Problem Solving in Mathematics Education

What problem solving and problem-solving strategies mean in school mathematics. Name the framework the thesis adopts (e.g. Pólya, Schoenfeld, NCTM process standards) and verify each source before citing it.

1.2.2 Senior Secondary Mathematics in Sri Lanka

G.C.E. A/L mathematics streams, the curriculum and competency-based aims (NIE), examination performance evidence (Department of Examinations). Proposal §1.1 cites DOE 2018/2019 and NIE 2019; these are not yet in the reference list (see analysis.typ § Citation audit).

1.2.3 Teachers' Competency in Using Problem-Solving Strategies

Evidence on teachers' content and pedagogical knowledge (Proposal §1.1: ICMI 2016, Siregar 2017, NEC 2018). Say what "competency" includes in this study.

1.3 Statement of the Problem

Objectives I, II, III

Condense Proposal §1.2 into a problem statement: the gap between teachers' subject knowledge and their use of problem-solving strategies in class, and the effect on students. End with the gap this study addresses.

1.4 Purpose of the Study

One paragraph linking the problem to the three phases and the teacher education module.

1.5 Research Objectives

The objectives of the study are to:

- I. Identify the current level of cognitive abilities of secondary mathematics teachers regarding problem-solving strategies.
- II. Explore the level of use of problem solving strategies in mathematics by senior secondary teachers.
- III. Find out areas for improvement in problem-solving strategy among senior secondary teachers.
- IV. Design and implement an intervention to enhance teachers' competency in applying problem-solving strategies in mathematics.
- V. Assess the progress of use of problem-solving strategies in mathematics after intervention.
- VI. Propose a teacher education module on effective use of problem solving strategies in mathematics in senior secondary education.

1.6 Research Questions

Objectives I, II, III, IV, V, VI

The proposal lists no research questions, but §4.1 refers to them. Write at least one question per objective (I–VI), numbered RQ1–RQ6.

1.7 Hypotheses

Objectives I, II, V

Null and alternative hypotheses for the Phase 1 relationships and for the Phase 3 pre/post comparison (H_0 ...). Leave this out if the design is not hypothesis-testing.

1.8 Significance of the Study

Expected value for teachers, teacher educators, NIE, zonal education offices and policy.
Source: Proposal §5 (Potential contributions).

1.9 Conceptual Framework

Objectives I, II, III, IV, V, VI

Diagram and explanation of the study's constructs and how they link: cognitive ability → use of strategies → areas for improvement → intervention → progress → module. Resolve the construct drift between competency and attitudes/beliefs noted in analysis. typ § Construct alignment.

Conceptual framework diagram (candidate to design)

Figure 1.1: Conceptual framework of the study

1.10 Definition of Terms

Operational definitions of: problem-solving strategy; competency; cognitive ability (Objective I); level of use (Objective II); senior secondary teacher; intervention; teacher education module.

1.11 Scope and Delimitations

Population (G.C.E. A/L mathematics teachers), education zones covered, media of instruction (Sinhala, Tamil, English), period of study.

1.12 Limitations of the Study

Constraints on sampling, access, time and generalisability.

1.13 Ethical Considerations

Brief summary: UoC Faculty of Education ethics clearance, informed consent, voluntary participation, pseudonyms (T01, S01, Z01). Full detail goes in §3.

1.14 Organisation of the Thesis

One short paragraph per chapter (1–5).

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Scope of the review, search strategy (databases, keywords, years), and how the chapter is organised. Proposal §2 names four themes: theoretical framework, previous studies, problem-solving strategies, Sri Lankan context.

2.2 Theoretical Framework

Objectives I, II, IV

2.2.1 Theories of Mathematical Problem Solving

E.g. Pólya's four stages; Schoenfeld's framework (resources, heuristics, control, beliefs). Say which one the instruments and intervention will use.

2.2.2 Teacher Knowledge and Competency

E.g. Mathematical Knowledge for Teaching (Ball, Thames & Phelps, 2008), verified before citing. Link it to Objective I (cognitive ability).

2.2.3 Teachers' Attitudes, Beliefs and Self-Efficacy

Proposal §2 names the Theory of Planned Behaviour and Self-Efficacy Theory but gives no sources. Keep this only if these constructs are measured. State how they relate to competency.

2.2.4 Learning Theories Underpinning the Intervention

E.g. Vygotsky's social development theory (Proposal §2 table; not yet in the reference list), collaborative problem solving.

2.3 Problem-Solving Strategies in Secondary Mathematics

Objective II

Typology of strategies (heuristics, e.g. working backwards, drawing a diagram, finding a pattern, generalising), and evidence of their effectiveness at secondary level.

2.4 Teachers' Use of Problem-Solving Strategies

Objectives I, II, III

2.4.1 International Studies

What the research says about teachers' competency and classroom use.

2.4.2 Studies in Sri Lanka and South Asia

Local evidence (NIE, NEC, DOE reports; university theses).

2.4.3 Barriers and Enabling Factors

Time, examination pressure, resources, training, beliefs.

2.5 Teacher Professional Development and Interventions

Objectives IV, V

Models of professional development, features of effective programmes (duration, collective participation, classroom practice, coaching), and how change in teachers' practice has been measured.

2.6 Teacher Education Modules

Objective VI

Design principles for in-service and pre-service modules; the Sri Lankan teacher education system (NIE, National Colleges of Education, universities, zonal in-service advisers).

2.7 The Sri Lankan Context

Objectives I, II

2.7.1 G.C.E. A/L Mathematics Curriculum

Competencies, syllabus, and teachers' guides (NIE).

2.7.2 Student Performance in Mathematics

G.C.E. A/L results evidence (Department of Examinations).

2.7.3 Mathematics Teacher Preparation and In-Service Training

Qualifications, SBTD programmes, in-service advisers.

2.8 Summary of Related Studies

Replace the rows with verified studies the candidate has read. Proposal §2 table rows (NIE 2019, DOE 2018, Vygotsky 1978, Lesh et al. 2019) must be verified first.

Table 2.1: Summary of related studies

Author (year)	Focus	Key finding	Relevance to this study
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

2.9 Research Gap

Objectives I, II, III, IV, V, VI

What is not yet known, and how each objective addresses it.

2.10 Summary

Key points carried into the methodology.

CHAPTER 3

METHODOLOGY

3.1 Introduction

Overview of the chapter and of the multi-phase mixed-methods design.

3.2 Research Paradigm

Philosophical stance (e.g. pragmatism) and why it fits a mixed-methods, intervention study.

3.3 Research Design

Objectives I, II, III, IV, V, VI

Multi-phase mixed-methods design (Proposal §4): Phase 1 quantitative survey → Phase 2 qualitative exploration → Phase 3 intervention → teacher education module. Explain how each phase informs the next.

Design diagram, redrawn from Proposal §4 (p. 8) after the sampling issues are resolved

Figure 3.1: Multi-phase mixed-methods research design

Table 3.1: Research phases and objectives, as stated in the proposal

Phase	Approach	Method	Objectives
Phase 1	Quantitative Phase	Cross-sectional survey	I, II
Phase 2	Qualitative Phase	Exploratory study	I, II, III
Phase 3	Quantitative & Qualitative mixed design using experimental design	Intervention (experimental / action component)	IV, V, VI

Phase 3 → IV, V, VI is inferred, not stated in the proposal. Confirm it.

3.4 Population

G.C.E. A/L mathematics teachers in Sri Lanka: number, distribution by province, zone and medium. Note that the proposal says “secondary” in Participants but “senior secondary” in the title.

3.5 Phase 1: Quantitative Survey

Objectives I, II

3.5.1 Sampling and Sample Size

One sampling frame and one N, with a justification (power analysis for regression; 5–10 respondents per item for factor analysis). The proposal gives 45 (stratified), 150 (convenience) and “purposive”; resolve to one.

Table 3.2: Phase 1 sample by education zone

Education zone	Schools	Teachers (population)	Teachers (sample)
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

3.5.2 Instrument: Teacher Questionnaire

Sections, number of items, response scale, and which items measure Objective I (cognitive ability) and which measure Objective II (level of use).

Table 3.3: Structure of the teacher questionnaire

Section	Construct	No. of items	Objective
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

3.5.3 Validity and Reliability

Expert panel and content validity index (CVI); pilot study (sample, setting); Cronbach's α ($\geq .70$) per scale; EFA/CFA if N allows.

3.5.4 Data Collection Procedure

Permissions (MOE, provincial and zonal offices), mode (online or in person), timing, follow-up.

3.5.5 Data Analysis

Software (R or Python), data screening, descriptive statistics, and the inferential tests for each research question. The proposal lists factor analysis, discriminant function analysis and regression.

3.6 Phase 2: Qualitative Exploration

Objectives I, II, III

3.6.1 Participants and Selection

How participants are drawn from Phase I (the proposal gives 45 and 30 interviews, and 20 observed classes). Selection criteria. Pseudonyms T01 ...

3.6.2 Instruments

Semi-structured interview schedule, classroom observation checklist, document analysis guide (lesson plans, schemes of work). Say how each was developed and piloted.

3.6.3 Data Collection Procedure

Interview setting, language, recording, transcription and translation; observation protocol.

3.6.4 Data Analysis

Reflexive thematic analysis (Braun & Clarke), coding in NVivo (the proposal names QSR N6, which is legacy software). Codebook development.

3.6.5 Trustworthiness

Credibility, transferability, dependability, confirmability; triangulation and member checking.

3.7 Phase 3: Intervention

Objectives IV, V

3.7.1 Design

Quasi-experimental pre/post design, with a comparison group if feasible. The proposal has no control group or pre/post instrument.

3.7.2 Participants

Sampling (the proposal says “convenient”), group sizes, assignment.

3.7.3 The Intervention

How Phase 1–2 findings shaped it; aims, content, duration, sessions, delivery mode, facilitators, materials.

Table 3.4: Intervention plan

Session	Focus / strategy	Activities	Duration
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

3.7.4 Fidelity of Implementation

Attendance, session checklists, observation and reflection logs.

3.7.5 Instruments and Measures

Pre/post measures of competency in using problem-solving strategies; classroom observation; teacher reflections; stakeholder feedback.

3.7.6 Data Analysis

Pre/post comparison (e.g. paired t-test or Wilcoxon; ANCOVA with a comparison group), effect sizes, qualitative analysis of reflections.

3.8 Development of the Teacher Education Module

Objective VI

The proposal has no method for Objective VI. Describe the design model, how findings feed into it, expert review and validation of the module.

3.9 Integration of Quantitative and Qualitative Data

Points of integration between phases; joint displays; meta-inferences.

3.10 Language of Instruments

Sinhala, Tamil and English versions; translation and back-translation procedure; checking equivalence.

3.11 Ethical Considerations

UoC Faculty of Education ethics clearance; permissions; information sheet and consent; the right to withdraw; confidentiality and pseudonyms; data storage; no identifiable data in AI tools.

3.12 Summary

A table or paragraph mapping objective $\rightarrow$ research question $\rightarrow$ phase $\rightarrow$ instrument $\rightarrow$ analysis.

Table 3.5: Alignment of objectives, questions, data and analysis

[illegible]

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

Objectives I, II, III, IV, V, VI

What this chapter reports and in what order: Phase 1 results, Phase 2 findings, Phase 3 results and the module, then the discussion.

4.2 Results of Phase 1: Quantitative Survey

4.2.1 Overview

Objectives I, II

What this chapter reports, and in what order (by research question).

4.2.2 Response Rate and Data Screening

Questionnaires distributed, returned and usable; missing data; outliers; normality checks.

4.2.3 Profile of Respondents

Table 4.1: Profile of respondents

Characteristic	Category	n	%
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Gender, age, qualifications, professional training, years of A/L teaching, education zone, school type, medium.

4.2.4 Validity and Reliability of the Instrument

Table 4.2: Reliability of questionnaire scales

Scale	No. of items	Cronbach's α	Retained items
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

EFA/CFA results if run: KMO, Bartlett's test, loadings, fit indices.

4.2.5 Teachers' Cognitive Abilities Regarding Problem-Solving Strategies

Objective I

Table 4.3: Descriptive statistics for cognitive ability

Dimension	M	SD	Level
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

Results only; interpretation goes in the Discussion (§4.5).

4.2.6 Teachers' Level of Use of Problem-Solving Strategies

Objective II

Table 4.4: Descriptive statistics for level of use

Strategy	M	SD	Level
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

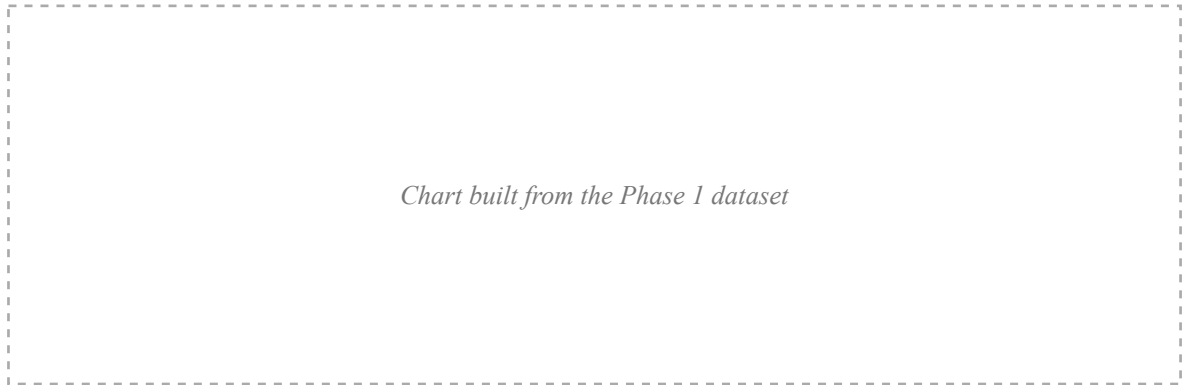


Figure 4.1: Level of use of problem-solving strategies

4.2.7 Differences by Teacher Characteristics

Objectives I, II

Group comparisons (e.g. by zone, experience, training), if they are among the research questions.

4.2.8 Relationships and Predictors

Objectives I, II

Table 4.5: Regression of level of use on predictors

Predictor	B	SE	β	p
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

Model summary (R^2 , F), assumptions checked.

4.2.9 Summary

Findings carried forward into Phase 2.

4.3 Findings of Phase 2: Qualitative Exploration

4.3.1 Overview

Objectives I, II, III

Purpose of Phase 2 and how it builds on Phase 1.

4.3.2 Profile of Participants

Table 4.6: Profile of Phase 2 participants

Pseudonym	Zone	School type	Years of A/L teaching	Data source
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

4.3.3 Analytic Process

Coding stages, codebook, and how themes were generated and reviewed.

Table 4.7: Themes and sub-themes

Theme	Sub-theme	Participants	Objective
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

4.3.4 Teachers' Understanding of Problem-Solving Strategies

Objective I

Themes on teachers' cognitive abilities, with anonymised quotations.

4.3.5 Teachers' Classroom Use of Problem-Solving Strategies

Objective II

4.3.5.1 Evidence from Interviews

Themes and quotations.

4.3.5.2 Evidence from Classroom Observation

Observation checklist results and field notes.

4.3.5.3 Evidence from Documents

Lesson plans, schemes of work, teachers' guides.

4.3.6 Challenges and Areas for Improvement

Objective III

Barriers, needs and areas for improvement identified by teachers and observed in class.

4.3.7 Integration with Phase 1 Results

Objectives I, II, III

Where the qualitative findings confirm, extend or contradict Phase 1.

Table 4.8: Joint display of Phase 1 and Phase 2 findings

Aspect	Phase 1 result	Phase 2 finding	Meta-inference
—	—	—	—
—	—	—	—
—	—	—	—

4.3.8 Summary

Findings that shape the intervention design in Phase 3.

4.4 Results of Phase 3: Intervention and Teacher Education Module

4.4.1 Overview

Objectives IV, V, VI

Purpose of Phase 3 and how Phases 1–2 shaped it.

4.4.2 Design of the Intervention

Objective IV

How the areas for improvement (Phase 2, §4.3) became the aims and content of the intervention; the final intervention plan.

Intervention flow diagram (candidate to design)

Figure 4.2: Structure of the intervention

4.4.3 Implementation of the Intervention

Objective IV

4.4.3.1 Participants and Setting

Table 4.9: Phase 3 participants

Group	n	Zone(s)	Schools
—	—	—	—
—	—	—	—
—	—	—	—

4.4.3.2 Delivery and Fidelity

Sessions delivered, attendance, deviations from the plan, fidelity checks.

4.4.4 Progress in the Use of Problem-Solving Strategies

Objective V

4.4.4.1 Pre-Intervention Results

Baseline measures.

4.4.4.2 Post-Intervention Results

Table 4.10: Pre- and post-intervention scores

Measure	Pre M (SD)	Post M (SD)	Test statistic	p	Effect size
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—

4.4.4.3 Classroom Observation after the Intervention

Changes observed in practice.

4.4.4.4 Teachers' Reflections and Stakeholder Feedback

Themes from reflections and feedback, with anonymised quotations.

4.4.5 Proposed Teacher Education Module

Objective VI

4.4.5.1 Rationale and Aims

Why the module is needed, drawing on all three phases.

4.4.5.2 Structure and Content

Table 4.11: Structure of the teacher education module

Unit	Learning outcomes	Content	Activities	Hours
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

4.4.5.3 Delivery and Assessment

Target users (pre-service or in-service), delivery mode, assessment of participants.

4.4.5.4 Validation of the Module

Expert review, pilot, revisions made.

4.4.6 Summary

Key results of Phase 3.

4.5 Discussion

4.5.1 Overview

How the discussion is organised: by objective, then integrated.

4.5.2 Teachers' Cognitive Abilities Regarding Problem-Solving Strategies

Objective I

Phase 1 and 2 findings set against the literature (Chapter 2).

4.5.3 Level of Use of Problem-Solving Strategies

Objective II

Discussion of Objective II findings.

4.5.4 Areas for Improvement

Objective III

Discussion of Objective III findings.

4.5.5 Effectiveness of the Intervention

Objectives IV, V

Discussion of Objectives IV and V.

4.5.6 The Teacher Education Module

Objective VI

How the module responds to the findings; comparison with existing programmes.

4.5.7 Integrated Discussion

Objectives I, II, III, IV, V, VI

Mixed-methods meta-inferences across the three phases.

4.5.8 Theoretical and Practical Contributions

What the study adds to the theory and practice of problem-solving teaching.

4.5.9 Summary

Key points of the discussion.

4.6 Summary

Key results and discussion points carried into Chapter 5.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

Purpose of the chapter.

5.2 Summary of the Study

Problem, objectives, design, sample and instruments in brief.

5.3 Major Findings

Objectives I, II, III, IV, V, VI

One subsection or paragraph per research question or objective.

5.4 Conclusions

What the findings mean in answer to each research question.

5.5 Recommendations

5.5.1 For Mathematics Teachers

Candidate to draft.

5.5.2 For Teacher Educators and Teacher Education Institutions

NIE, National Colleges of Education, universities, in-service advisers.

5.5.3 For Policy Makers

Ministry of Education, provincial and zonal education offices.

5.6 Contributions of the Study

Contribution to knowledge, practice and policy. Source: Proposal §5.

5.7 Limitations

Limitations that affect how far the findings can be generalised.

5.8 Suggestions for Further Research

Candidate to draft.

5.9 Summary

Closing paragraph.

REFERENCES

APPENDICES

Appendix-1: Ethics Clearance

UoC Faculty of Education ethics approval letter.

Appendix-2: Permission Letters

Ministry of Education, provincial and zonal education offices, principals.

Appendix-3: Participant Information Sheet and Consent Form

Blank templates only, in Sinhala, Tamil and English. No signed forms.

Appendix-4: Teacher Questionnaire (Phase 1)

Final version in each language; note the translation procedure.

Appendix-5: Content Validity: Expert Panel and CVI

Panel composition (anonymised) and item-level CVI.

Appendix-6: Interview Schedule (Phase 2)

Candidate to draft.

Appendix-7: Classroom Observation Checklist

Candidate to draft.

Appendix-8: Document Analysis Guide

Candidate to draft.

Appendix-9: Pre- and Post-Intervention Instruments (Phase 3)

Candidate to draft.

Appendix-10: Intervention Plan and Session Materials

Candidate to draft.

Appendix-11: Teacher Education Module

Full module document.